

Title:

Development of a simulation model for fly casting and application to overhead casting.

Authors

H. Ekander¹, N. C. Perkins², B. W. Richards³

Abstract

Fly casting is both a competitive sport and the fishing technique used in recreational fly fishing. This paper presents a physics-based simulation model for fly casting including the coupled dynamics of the fly rod, fly line, leader/tippet and fly. This model extends prior models by including material damping and Reynolds number dependent air drag, both improving accuracy of simulations. Experimental data describes the casting stroke applied by the caster's hand to the fly rod. Results demonstrate how variations in casting strokes and variations in tackle affect casting dynamics. The energy dynamics during the cast reveals the key influence of air drag and energy transfer between the rod and the fly line.

1 Introduction

Fly casting is a method of casting a light lure (fly) using a long, flexible fly rod and a weighted, plastic-coated fly line. The fly line is retained on the fly rod by small metal rings called guides. When casting a fly, the angler holds the rod and swings it largely in the vertical plane in a precise casting stroke to cast the weighted fly line. In conventional spin or bait casting the lure provides the casting mass and it pulls the line from the reel. In fly casting, the fly line provides the casting mass, and it forms what anglers refer to as a loop, Fig. 1d, which is the U-shape of the line travelling through the air, with the fly trailing behind, [1-4].

The fly line is tapered in specific ways to affect how the line travels through the air and delivers the fly to the intended target. The fly line is visible to the fish so a clear monofilament leader typically 2-4 meters long is attached to the end of the fly line to distance the visible fly line from the fly. The small diameter end of the leader that attaches to the fly is called the tippet and it is nearly invisible to the fish. Collectively, the gear used when fishing is referred to as tackle.

Fly casting has also developed into a competitive sport, controlled by the International Casting Sport Federation. Successful fly casting requires well-matched tackle components [5,6] and exceptionally well controlled casting strokes by the caster [1-4,7]. Fly casting using a fly rod with guides and a fly line has evolved from the mid-19th century. The opportunities provided by material developments have, over the years, led to efficient tackle. Modern stiff fibers have paved the way for light and fast fly rods [5,8,9] and polymer-based materials have led to improved fly lines [5,10,11]. Casting techniques adapted to modern tackle [1-3,7] have extended casting ranges and decreased the effort involved in casting. Developments of new tackle largely follow qualitative trial and error methods although some quantitative (e.g. model-based) methods exist, particularly for the fly rod [9] and fly line [10]. Proficient fly casting takes a long time to master and individuals evaluating tackle by casting necessarily rely on their individual "feel".

The dynamics of fly casting has been analyzed using models of just the fly line loop [10-13] as well as models of the coupled rod and line system [9]. These models capture/explain some aspects of a fly cast but fall short of capturing the complete dynamics. Simulation models have been developed using finite element methods [14] as well as finite difference methods [12] with results in close agreement with experiments. These models have many commonalities with models used in mechanical simulation of submarine cables and mooring systems [15-17].

This paper presents a physics-based simulation model for fly casting including the coupled dynamics of the fly rod, fly line, leader/tippet and fly. This model extends prior models by including material damping and Reynolds number dependent air drag, both improving accuracy of simulations. Such a quantitative model reveals how casting dynamics is affected by changes in tackle and by changes in the casting stroke. Predictions by the model are compared with practical experiences [1-4]. Furthermore, quantities essential for understanding the dynamics of fly casting that are difficult to measure (e.g., air drag, forces in the rod and fly line) may be studied quantitatively. Finally, this paper

also contributes a detailed evaluation of mechanical energy dynamics, allowing improved understanding of casting dynamics.

2 Method

2.1 Physical modeling

The computational space includes fly rod, fly line and leader, see Fig. 1c, and is divided into subdomains connected by internal boundary conditions, BCs. The equations presented in this section are applicable to all subdomains. Two frames of reference are used, an inertial frame x, y, z and a Lagrangian co-moving frame s, n, z , see Fig. 1a-b:

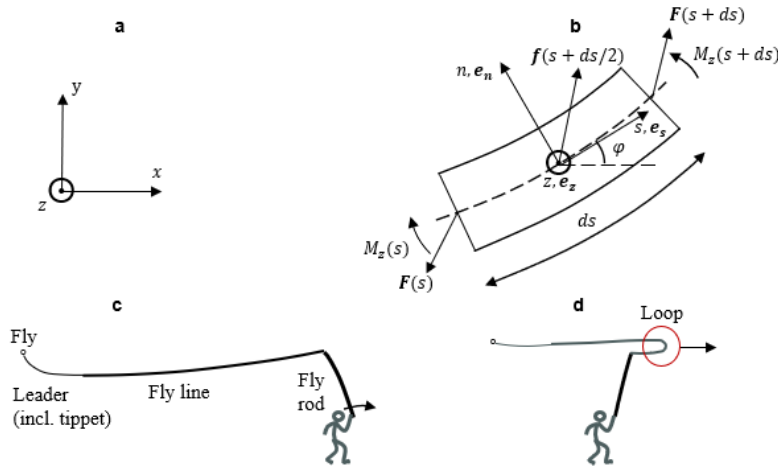


Fig. 1 Inertial frame of reference x, y, z **a**, and moving frame of reference s, n, z with unit vectors e_s, e_n, e_z and force vector definitions with F denoting the internal line/rod force (tensile and normal), M_z the internal bending moment resulting from the stiffness of the line/rod and f the net external force per unit length (incl. gravity and air drag) **b**, Terminology for tackle with rod rotation indicated by arrow **c**, “Loop” circled with arrow showing propagation **d**

2.1.1 Momentum equations

The two-dimensional equations in [16] are further developed here to model fly casting by treating the fly rod, the fly line, and the leader as (distinct but) continuous beams. This continuum modeling approach is consistent with some prior models of fly casting dynamics (e.g. [12,14]) yet distinct from others (e.g., [18]) employing lumped-parameter chains of links. With $\mathbf{u} = (u_s, u_n, 0)$ denoting the velocity, $\mathbf{v} = (0, 0, v_z = \partial\varphi/\partial s)$ the curvature, $\boldsymbol{\omega} = (0, 0, \partial\varphi/\partial t)$ the angular velocity and neglecting elongation, the kinematic conditions of continuity in space and time follows from [16]:

$$\frac{\partial u_s}{\partial s} = u_n \cdot v_z \quad (1)$$

$$\frac{\partial \varphi}{\partial t} - \frac{\partial u_n}{\partial s} = u_s \cdot v_z \quad (2)$$

With $\mathbf{F} = (F_s, F_n, 0)$, m denoting the mass per unit length, g the constant of gravity and neglecting the added mass included in [16], the equations of motion for an infinitesimal element of flyrod/fly line/leader in the tangential and normal directions may be written as follows using equations (1,2):

$$m \cdot \left(\frac{\partial u_s}{\partial t} - u_n \cdot \frac{\partial u_n}{\partial s} - u_s \cdot \frac{\partial u_s}{\partial s} \right) = \frac{\partial F_s}{\partial s} - v_z \cdot F_n - m \cdot g \cdot \sin(\varphi) - f_{s,drag} \quad (3)$$

$$m \cdot \left(\frac{\partial u_n}{\partial t} + u_s \cdot \frac{\partial u_n}{\partial s} + u_s^2 \cdot \frac{\partial \varphi}{\partial s} \right) = \frac{\partial F_n}{\partial s} + v_z \cdot F_s - m \cdot g \cdot \cos(\varphi) - f_{n,drag} \quad (4)$$

Here, the left hand sides of equations (3) and (4) represent the mass per unit length times the acceleration in the tangential and normal directions, respectively. The right-hand sides represent the net force per unit length in the tangential and normal directions, respectively, including the internal tangential (or tension) force differential $dF_s/ds = \partial F_s/\partial s - v_z \cdot F_n$, the internal normal (or shear) force differential $dF_n/ds = \partial F_n/\partial s + v_z \cdot F_s$, the gravity force per unit length, and air drag force per unit length. In particular, the air drag forces (including form- and friction drag) per unit length $f_{s,drag}, f_{n,drag}$ depend on Reynolds number, Re . $Re = \rho_{air} \cdot u \cdot d / \mu_{air}$ where u, d denote characteristic velocity and diameter, respectively and $\rho_{air} \approx 1.2 \text{ kg/m}^3$, $\mu_{air} \approx 1.8 \cdot 10^{-5} \text{ Ns/m}^2$ denote density and dynamic viscosity of air, respectively.

In fly casting situations, the Reynolds number based on diameter d varies from the order 1 (velocity 0.1 m/s and diameter $0.2 \cdot 10^{-3} \text{ m}$ at the end of the leader) to the order 10^3 (velocity 25 m/s and diameter $1.2 \cdot 10^{-3} \text{ m}$ for the fly line). A correlation is applied with the constants, $\zeta_{n,0} = 1.0$, $\zeta_n = 10$ and $m_n = -0.75$, agreeing with experimental data [19] within about 10% on the range $1 < Re_{n,d} < 2000$:

$$f_{n,drag} = \zeta_{n,0} \cdot d \cdot \frac{\rho_{air} \cdot |u_n| \cdot u_n}{2} + \zeta_n \cdot \left(\frac{\rho_{air} \cdot d}{\mu_{air}} \right)^{m_n} \cdot d \cdot 0.5 \cdot \rho_{air} \cdot |u_n|^{1+m_n} \cdot u_n \quad (5)$$

A similar equation with the constants: $\zeta_{s,0} = 0.0022$, $\zeta_s = 2.0$ and $m_s = -0.91$ is applied for the drag force in the s -direction:

$$f_{s,drag} = \zeta_{s,0} \cdot \pi \cdot d \cdot \frac{\rho_{air}}{2} \cdot |u_s| \cdot u_s + \zeta_s \cdot \left(\frac{\rho_{air} \cdot d}{\mu_{air}} \right)^{m_s} \cdot \pi \cdot d \cdot 0.5 \cdot \rho_{air} \cdot |u_s|^{1+m_s} \cdot u_s \quad (6)$$

Euler-Bernoulli beam theory [20] is applied in [16] using a linear elastic material model, $\sigma = E \cdot \varepsilon$ (stress equals Young's modulus times strain). However, for the materials considered here (the fly line in particular), energy is dissipated when the strain changes with time. This phenomenon is referred to as material damping and is modeled using a Kelvin-Voigt viscoelastic model [20], $\sigma = E \cdot (\varepsilon + \eta \cdot d\varepsilon/dt)$ with η denoting the material relaxation time. The viscous term is comparatively small and is calculated with $d\varepsilon/dt = \partial\varepsilon/\partial t$. Applying the viscoelastic model, the relation between bending moment M_z and curvature $v_z = \partial\varphi/\partial s$ follows (I denoting the second area moment of inertia):

$$M_z \approx EI \cdot \left(v_z + \eta \cdot \frac{\partial v_z}{\partial t} \right) \quad (7)$$

For a slender body, the rotational inertia of the cross section is small and may be neglected [16] giving the following relation between normal force, and bending moment:

$$\frac{\partial M_z}{\partial s} = -F_n \quad (8)$$

With $EI = EI(s)$ and η constant on each subdomain, equations (7,8) give:

$$EI \cdot \left(\Gamma_z + \eta \cdot \frac{\partial \Gamma_z}{\partial t} \right) + \frac{\partial}{\partial s}(EI) \cdot \left(v_z + \eta \cdot \frac{\partial v_z}{\partial t} \right) = -F_n \quad (9)$$

With $v_z = \partial\varphi/\partial s$ and $\Gamma_z = \partial v_z/\partial s$ equations (1), (2), (3 with 6), (4 with 5) and (9) form a set of seven first order equations for the seven variables $u_s, u_n, F_s, F_n, \varphi, v_z, \Gamma_z$. The set of equations is solved with internal and external BCs.

2.1.2 Boundary conditions

The variables computed on the starting and ending s -boundaries of a subdomain are denoted $()^-$ and $()^+$ respectively. At an internal boundary (e.g., between fly line and leader and between the rod and the fly line), all solution variables except v_z and Γ_z are continuous. Continuous bending moment and normal force across internal boundaries with $\partial(EI)/\partial s = 0$ give the following internal BCs for the solution variables v_z and Γ_z :

$$\left(EI \cdot \left(v_z + \eta \cdot \frac{\partial v_z}{\partial t} \right) \right)^- = \left(EI \cdot \left(v_z + \eta \cdot \frac{\partial v_z}{\partial t} \right) \right)^+ \quad (10)$$

$$\left(EI \cdot \left(\Gamma_z + \eta \cdot \frac{\partial \Gamma_z}{\partial t} \right) \right)^- = \left(EI \cdot \left(\Gamma_z + \eta \cdot \frac{\partial \Gamma_z}{\partial t} \right) \right)^+ \quad (11)$$

External BCs are required to form a closed problem. The velocities u_s, u_n and the angle φ are prescribed as functions of time at the rod handle, see section 2.3. A fly is attached at the terminal end of the computational domain, i.e., at the end of the leader. The fly is considered small and spherically symmetric giving zero bending moment $(M_z)^+$ on the leader as one terminal BC. The equations of motion for a spherical fly with mass m_{fly} and diameter d_{fly} provide the additional terminal BCs for the forces $(F_s)^+, (F_n)^+$:

$$m_{fly} \cdot \left(\frac{\partial u_s}{\partial t} - u_n \cdot \frac{\partial \varphi}{\partial t} \right)^+ = - \left(\begin{array}{c} (F_s)^+ + m_{fly} \cdot g \cdot \sin(\varphi) + \\ \zeta_{fly,0} \cdot \frac{\pi \cdot d_{fly}^2 \cdot \rho_{air} \cdot |u_s| \cdot u_s}{4} + \\ \zeta_{fly} \cdot \frac{\pi \cdot d_{fly} \cdot \mu_{air} \cdot u_s}{8} \end{array} \right) \quad (12)$$

$$m_{fly} \cdot \left(\frac{\partial u_n}{\partial t} + u_s \cdot \frac{\partial \varphi}{\partial t} \right)^+ = - \left(\begin{array}{c} (F_n)^+ + m_{fly} \cdot g \cdot \cos(\varphi) + \\ \zeta_{fly,0} \cdot \frac{\pi \cdot d_{fly}^2 \cdot \rho_{air} \cdot |u_n| \cdot u_n}{4} + \\ \zeta_{fly} \cdot \frac{\pi \cdot d_{fly} \cdot \mu_{air} \cdot |u_n|}{8} \end{array} \right) \quad (13)$$

The air drag on the fly is calculated in analogy with equation (5), using the constants $\zeta_{fly,0} = 0.4$, $\zeta_{fly} = 55$, fitted to experimental data [19] for a sphere in the range $100 < Re_d < 20000$.

2.1.3 Energies

Energy dynamics during a fly cast is of interest, see sections 3.1 and 4. The kinetic, potential and strain energies $W_{kin}, W_{pot}, W_{strain}$ are integrated on each subdomain, i from $(s_i)^-$ to $(s_i)^+$, as follows [20]:

$$W_{kin,i}(t) = \frac{1}{2} \cdot \int_{(s_i)^-}^{(s_i)^+} m \cdot (u_s^2 + u_n^2) \cdot ds \quad (14)$$

$$W_{pot,i}(t) = g \cdot \int_{(s_i)^-}^{(s_i)^+} m \cdot y \cdot ds \quad (15)$$

$$W_{strain,i}(t) = \frac{1}{2} \cdot \int_{(s_i)^-}^{(s_i)^+} EI \cdot v_z^2 \cdot ds \quad (16)$$

Also, the dissipated energy by air drag, W_{drag} and material damping, W_{Mtrl} are integrated as is input work by the caster, W_{input} :

$$W_{drag,i}(t) = \int_0^t \int_{(s_i)^-}^{(s_i)^+} (f_{s,drag} \cdot u_s + f_{n,drag} \cdot u_n) \cdot ds \cdot dt \quad (17)$$

$$W_{Mtrl,i}(t) = \eta \cdot \int_0^t \int_{(s_i)^-}^{(s_i)^+} EI \cdot \left(\frac{\partial v_z}{\partial t} \right)^2 \cdot ds \cdot dt \quad (18)$$

$$W_{input}(t) = - \int_0^{(\varphi)^-} (M_z \cdot d\varphi)^- - \int_0^t ((F_s \cdot u_s + F_n \cdot u_n))^- \cdot dt \quad (19)$$

Here, the complete system includes the fly rod, fly line, leader, and fly and “line-system” includes the complete system excluding the rod. Denoting the difference in energy between $t = t$ and $t = 0$ by Δ , the conservation of mechanical energy is written:

$$\sum_i \left(\Delta W_{Kin,i} + \Delta W_{Pot,i} + \Delta W_{Strain,i} + W_{Mtrl,i}(t) + W_{Drag,i}(t) \right) = W_{Input}(t) \quad (20)$$

2.2 Numerics

The discretization in space is second order accurate and follows [16] and the discretization in time is based on the generalized alpha method, [21,22] which gives second order accuracy also in time. The implementation of the equations and the numerical integration has been verified through comparisons with exact solutions and examples are supplied as Online Resource 1.

The discrete equations are solved iteratively at each time step using the solution vector, $\mathbf{X}_j$ from the previous time step as the start solution for the next time step. The error vector $\mathbf{E}_k$ is computed and the Jacobian matrix is derived by differentiation with respect to the solution vector $\mathbf{J}_{k,j} = \partial \mathbf{E}_k / \partial \mathbf{X}_j$. The Jacobian matrix is inverted $\mathbf{J}_{k,j}^{-1}$ and used to update the solution vector at each iteration with $d\mathbf{X}_j = -\mathbf{J}_{k,j}^{-1} \mathbf{E}_k$ and the iterations are continued until the error norm $\|\mathbf{E}_k\|$ satisfies the convergence criteria, $\|\mathbf{E}_k\| < \varepsilon_{limit}$.

2.3 Cast description

An overhead cast [1-3] is composed of a forward and a back stroke. Here, the forward stroke is a mirror version of the back stroke giving symmetric casts. All simulated casts are obtained after several casting cycles (back and forward strokes) sufficient to produce a periodic (repeatable) solution.

The arc of a cast is defined as the maximum angular difference at the rod handle during a single (forward or back) casting stroke. A casting stroke is divided into four phases: acceleration, $0.26 < t < 0.68$ s, deceleration, $0.68 < t < 0.81$ s, rebound, $0.81 < t < 0.97$ s and pause $0.97 < t < 1.52$ s with the approximate times referring to the 66 deg arc cast measured for an expert caster studied in [4], see Fig. 4a. The casting stroke constitutes the BC at the rod handle and is defined by the angle and the translational velocities. Here, the casting strokes are based on angular velocity measured for an expert cast [4] performed using a 9.45 m fly line including leader and a 2.74 m long fly rod giving a casting arc of about 66 deg. Note that the expert cast reported in [4] used 7.62 m of flyline and leader (instead of the 9.45m of flyline + leader length herein), but was otherwise performed by the same expert and measured using the same method reported in [4]. The measured angular velocity is fitted to second order polynomials giving smoothed angular BC $(\phi_{sm}(t))^-$, see Fig. 4a. The BCs for the translational velocities, $(u_s(t))^-$, $(u_n(t))^-$ are set to simulate a cast made with the caster’s elbow kept fixed and with the wrist and the forearm allocated 75% and 25% of the angular motion respectively. The length from the elbow to the rod (excluding the handle) is 0.45 m.

2.4 Tackle description

Modern tackle used in fly casting is described in [5]. In the simulation model, the fly rod, fly line and leader are defined by m, d, EI, η as functions of s . The rod properties, Fig. 2, have been determined experimentally on a 4-piece, 2.74 m (2.47 m without handle), 5 weight fly rod. Applying weights and measuring deflections on different parts of the rod made it possible to determine $EI(s)$ and the mass distribution $m(s)$ was determined by weighing the rod parts. The relaxation time, $\eta \approx 0.0025$ s was determined by measuring the decay in deflection amplitude for the rod vibrating in the first eigenmode with the handle fixed.

The fly line properties are set to represent a 5 weight double tapered floating line, see Fig. 2, with a relaxation time of 0.1 s. It is noted that neither $EI(s)$ nor $EI(s) \cdot \eta$ have been determined by direct measurement so these values are approximate, section 4. The properties of leader/tippet are calculated based on $d(s)$ using a Young’s modulus of $3 \cdot 10^9$ N/m² and a density of 1200 kg/m³ representative for polyamide and the relaxation time is set to 0.3 s. A dry fly is represented by the mass $3 \cdot 10^{-5}$ kg and an equivalent sphere diameter of 0.016 m giving a terminal velocity at free fall of 2.4 m/s.

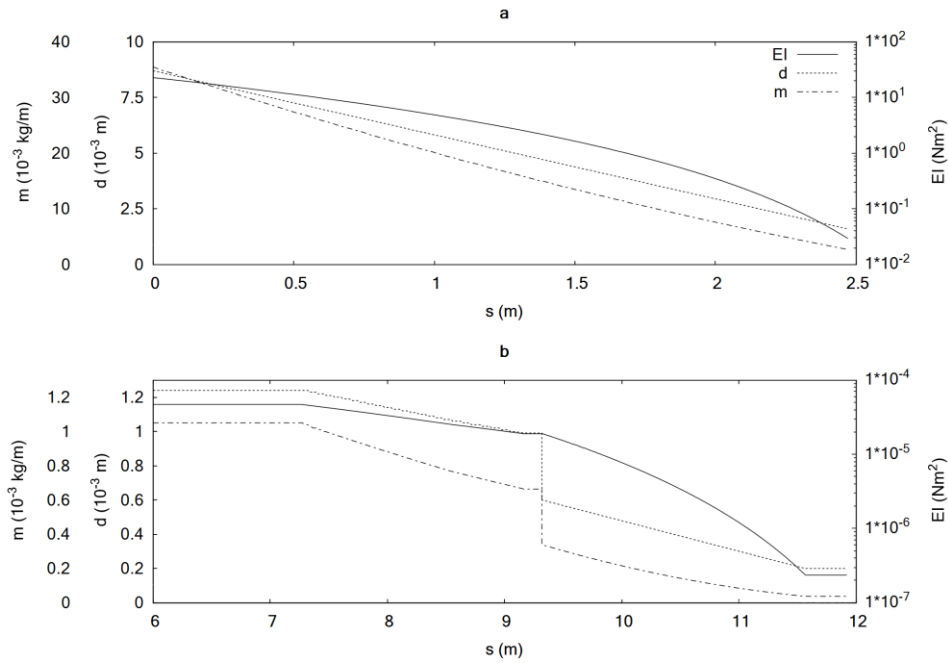


Fig. 2 Physical properties versus s-coordinate for fly rod **a**, and fly line with leader **b**. The fly line data is constant for $2.47 \leq s \leq 7.26$ m. The leader starts at $s = 9.32$ m and the tippet starts at $s = 11.57$ m. Note that $m(s)$ and $d(s)$ are plotted on the two left axes and $EI(s)$ on the right y-axis using a log-scale

3 Results

This section presents results from changes in how the cast is performed, section 3.1, and results from changing the tackle, sections 3.2 and 3.3. In addition, the energy dynamics during a cast is presented in section 3.1. Simulations reported here for casts, including a sequential forward and back stroke, are computed using about 1000 discrete points in space and 1500 points in time. The average number of iterations per time step is around 3 and the execution time is 15-18 s on a 2.7 GHz laptop. Animations of all casts are supplied as Online Resource 2.

3.1 Casting arc variation

In this section the casting arc is varied. Results for casting arcs 81, 66 and 51 deg, with the maximum angular velocity held almost constant, are presented in Fig. 3 and 4.

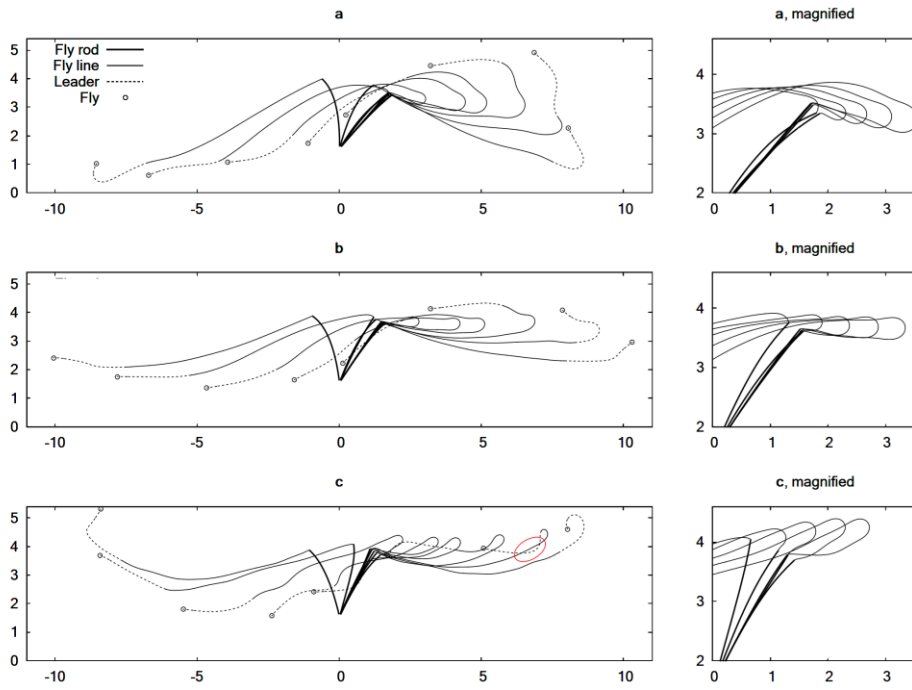


Fig. 3 Geometry (axes in m) for casting arcs 81 **a**, 66 **b** and 51 deg **c**, at equidistant times with time increasing as loop propagates to the right. Left graphs show 8 equidistant times of one stroke. Magnified graphs show loop formation at 5 equidistant times separated by $4/63$ of a stroke. Tailing loop circled in **c**

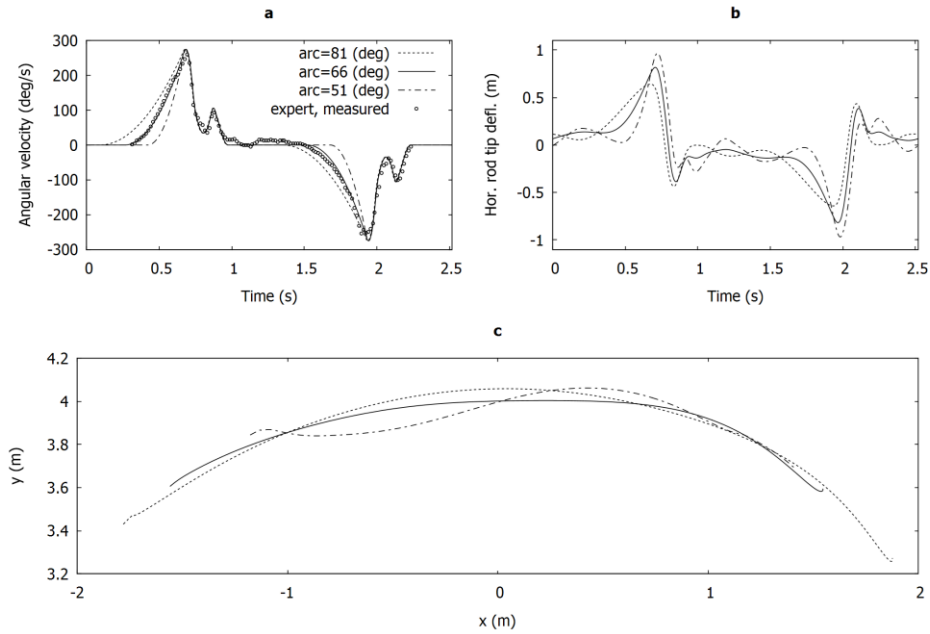


Fig. 4 Data for casting arcs 81, 66 and 51 degrees. Angular velocity at rod handle, BC and measured for expert cast [4], versus time for one cast including back and forward strokes **a**. Rod tip deflection i.e., horizontal difference between actual rod tip position and the position of a fictitious non-flexing rod **b**. Trajectory as rod tip moves from left to right in the graph during the stroke, about from 1.5 s to 2.3 s **c**

Energies, equations (14-16), for the complete system and the rod during half the 66 deg arc cast are presented in Fig. 5, as is the energy dissipation and input work, equations (17-19). The results are detailed in table 1 for the acceleration and acceleration + deceleration phases.

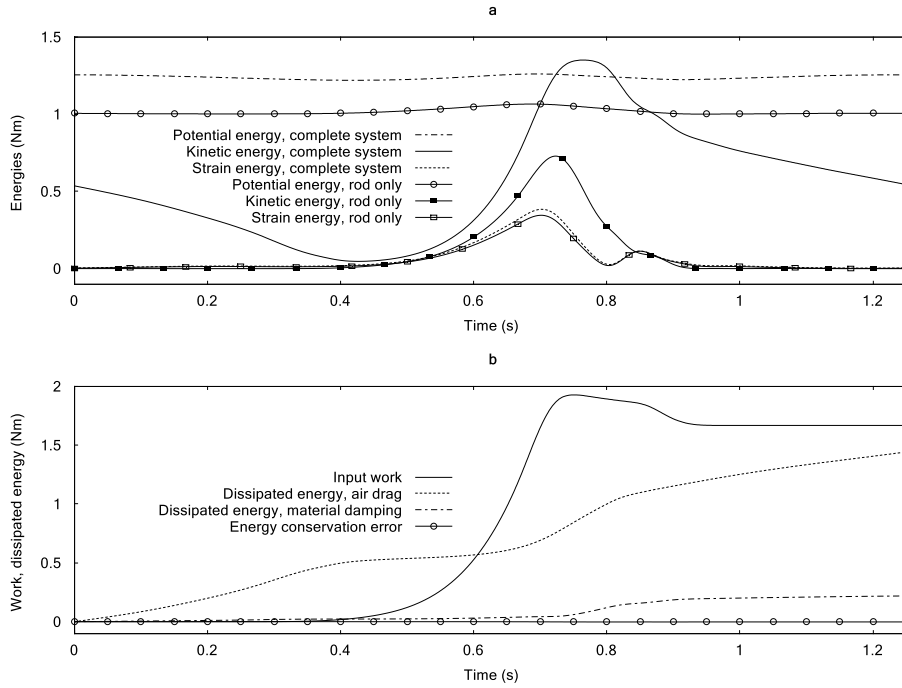


Fig. 5 Energies for the complete system and rod only **a**, dissipated energy for complete system, input work and conservation error denoting difference between the left- and right-hand sides of equation (20) **b**, presented for one stroke of the 66 deg arc cast

	Acceleration, $0.26 < t < 0.68$ s		Acceleration + deceleration, $0.26 < t < 0.81$ s	
	Rod	Line-system	Rod	Line-system
Energy dissipation, air drag	7%	20%	20%	19%
Energy dissipation, material damping	0%	2%	2%	4%
Change, kinetic energy	40%	6%	13%	41%
Change, strain energy	21%	2%	0%	0%
Change, potential energy	5%	-3%	2%	-1%

Table 1 Relative changes and dissipation of energy, for the 66 deg arc cast, during the acceleration and the acceleration + deceleration phases made non-dimensional using W_{input} for the respective phases

3.2 Rod stiffness variation

In this section the rod bending stiffness, EI_{rod} is varied. The rod bending stiffness presented in Fig. 2 is denoted $EI_{rod} = 100\%$. Proportionally decreased and increased rod bending stiffnesses are denoted $EI_{rod} = 60\%$ and $EI_{rod} = 167\%$ respectively. The rod mass per unit length is changed in proportion to the bending stiffness giving unchanged eigenfrequency for low amplitude rod vibrations. The casting strokes i.e., the BCs at the rod handle, correspond to the 66 deg arc cast in section 3.1. Results are presented in Fig. 6 and 7.

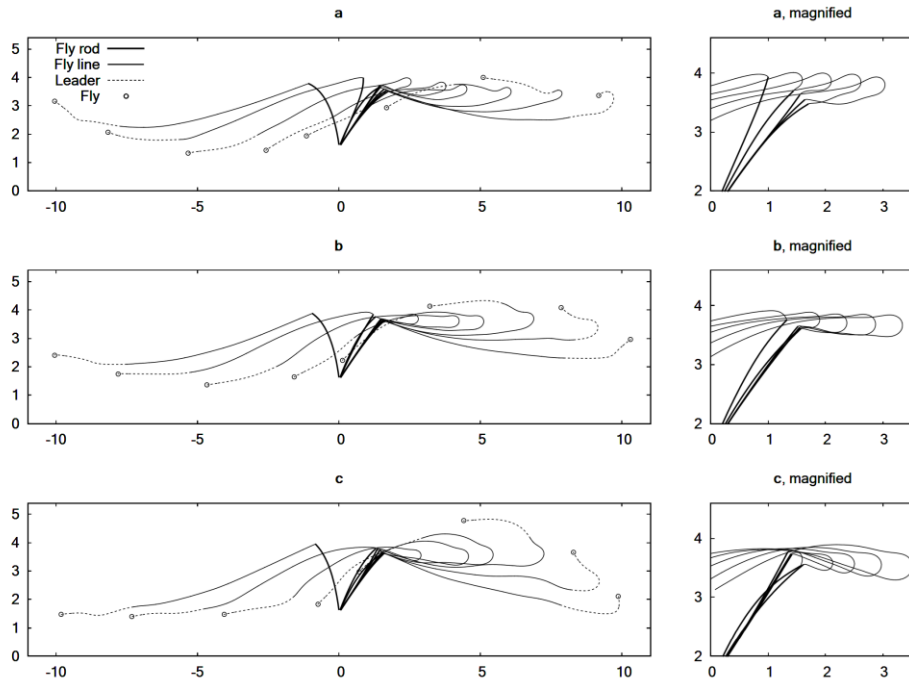


Fig. 6 Geometry (axes in m) for $El_{rod} = 60\%$ **a**, 100% **b** and 167% **c**, at equidistant times with time increasing as loop propagates to the right. Left graphs show 8 equidistant times of one stroke. Magnified graphs show loop formation at 5 equidistant times separated by $4/63$ of a stroke

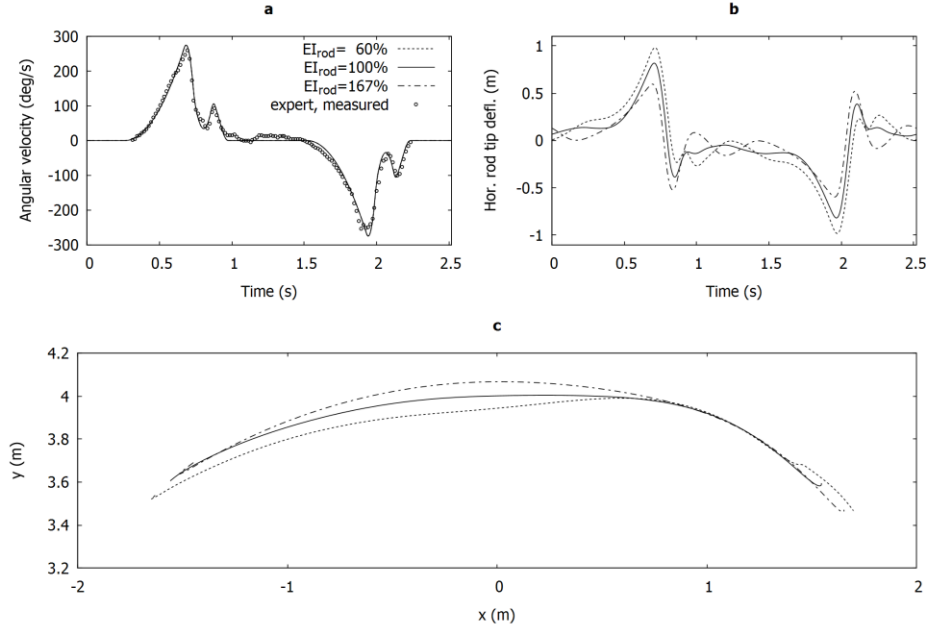


Fig. 7 Data for $EI_{rod} = 60\%$, 100% and 167% . Angular velocity BC at rod handle and measured for expert cast [4] versus time **a**. Rod tip deflection i.e., horizontal difference between actual rod tip position and the position of a fictitious non-flexing rod **b**. Trajectory as rod tip moves from left to right in the graph during the stroke, approximately from 1.5 s to 2.3 s **c**

3.3 Varying leader configuration

The purpose of varying the leader is to show the impact on the overturning of the fly line, leader and fly as the line is straightened. The leader presented in Fig. 2 is denoted a tapered leader. Two alternative leaders with constant diameters of $0.2 \cdot 10^{-3} \text{ m}$ and $0.6 \cdot 10^{-3} \text{ m}$ are denoted a thin and a thick leader respectively. The casting strokes i.e., the BCs at rod handle, are identical to the 66 deg arc cast in section 3.1. The results are presented in Fig. 8.

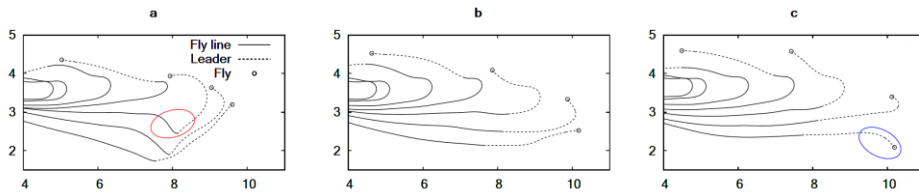


Fig. 8 Geometry (axes in m) for thin **a**, tapered **b** and thick leader **c**, at equidistant times with time increasing as loop propagates to the right. Division between curves is $1/24$ th of a cast. Fly line “overshoot” circled for thin leader in **a**. Leader “overshoot” circled for thick leader in **c**

4 Discussion

The results presented in Fig. 3 and 4 show the impact on loop formation by varying the casting arc. The loop size varies considerably, in agreement with practical experience [1-3]. In particular, the size and therefore the air drag on the loop is influenced by the path of the rod tip and straighter tip paths are known to yield smaller loops [1-3] with

less drag. Specifically, the wide loop for the 81 deg casting arc is associated with a large vertical rod tip movement, Fig. 4c. The air drag on the wide loop is large enough to slow down the line-system preventing a smooth overturn of the leader and fly at the end of the presented sequence. The 66 deg casting arc (angular velocity approximating an expert cast [4]) produces a low air drag, narrow loop as is expected, Fig. 3b. The overturning at the end of the shown sequence is smooth making the line-system almost straight when power is again applied at the start of the consecutive stroke, also in agreement with what is expected for an expert cast [4]. The 51 deg cast casting arc shows the formation of a “tailing” loop i.e., the undesirable crossing of the line with itself, as circled in Fig. 3c. The excessive rod loading for this cast makes the rod-tip dip before it starts moving up, see Fig. 4c, as is observed with tailing loops [1-3].

The rod bending stiffness influences the rod tip path as well as the loop shape, in agreement with practical experience [1-3]. Specifically, increasing EI_{rod} decreases the rod tip deflection, Fig. 7b and increases the loop-speed as shown by the loop-positions, Fig. 6a-c. The wider loop for the $EI_{rod} = 167\%$ compared to the $EI_{rod} = 100\%$, Fig. 6b-c, is associated with a larger vertical rod tip movement shown in Fig. 7c. The $EI_{rod} = 60\%$ rod produces a loop-shape resembling the one obtained for the 51 deg arc, Fig. 3. This is expected since the rods in both cases are excessively loaded. However, the $EI_{rod} = 60\%$ cast does not generate sufficient overload to produce a dip in the rod tip trajectory or a tailing loop.

The results presented in Fig. 8, obtained for the different leader configurations, show the impact on the line-system turnover prior to acceleration at the start of the consecutive casting stroke. The thin leader produces an “overshoot” of the fly line, circled in red, due to an insufficient mass*acceleration to normal drag ratio of the leader. The thick leader, on the other hand, produces an “overshoot” for the leader, circled in blue, due to an excessive mass*acceleration to normal drag ratio close to the fly.

The results presented in Fig. 5 and Table 1 show that during the acceleration phase the relative energy change of the rod is 66% and of the line-system 5%. The corresponding figures for the time including the deceleration phase are 15% and 40% demonstrating the transfer of strain and kinetic energy from the rod to the line-system during the deceleration phase. The change in kinetic energy for the line-system during acceleration and deceleration is 41% while the changes in other energies are <1%. This indicates that the dynamics for the line-system during the casting stroke is dominated by inertia compared to gravity and bending. The results in Table 1 also show that the energy dissipation is dominated by air drag, being 6-13 times larger than the material damping during the acceleration and deceleration phases. The relatively small contributions from material damping and changes in strain energy for the line-system suggest that the sensitivity of the solutions to changes in $EI(s)$ and $EI(s) \cdot \eta$ will be limited, see section 2.4.

The rebound is associated with the counter-flexing of the rod following the deceleration phase. The counter-flex produces the smaller local maximum in the angular velocity, see Fig. 4a. The input data for the rebound part of the casting stroke is measured for the 66 deg arc cast and this measured rebound has been applied also to the 81 and the 51 deg arc casts. It seems plausible, however, that the rebound may change if the cast is changed since the rebound is understood to be the result of dynamic interaction of the hand and the rod flexing after it has been decelerated.

The input work, equation (19), quantifies the transfer of energy across the boundary between caster and rod. The input work increases with time when energy is added to the rod and it decreases when energy is removed from the rod e.g., during the deceleration and rebound $0.68 < t < 0.97 \text{ s}$, see Fig. 5b. During the deceleration and rebound, the caster's muscles are working to stop the rod and to absorb rod oscillations. Hence, the caster's muscles are always doing positive work regardless of whether the input work is increasing or decreasing.

A real fly, partly made of soft materials, will change shape as speed increases making an accurate modeling of the fly air drag complicated. Here, the modeling of the fly is simplified by assuming constant properties but non-negligible drag.

The simulation model may be developed further e.g., to incorporate double haul. Double haul is a term used to describe hauling of line into the rod during a casting stroke and a later release of line from the rod. Double haul is applied in fly fishing and in competitive fly casting to increase rod loading and casting distance. Developments may also include interaction of the line with a water surface, hereby facilitating simulations of complete casts starting

with the line at water, lifting of the line from the water, false casts, delivery stroke and ending with the line having landed on the water surface.

5 Conclusions

This paper presents a physics-based simulation model for fly casting including the coupled dynamics of the fly rod, fly line, leader/tippet and fly. This model extends prior models by including material damping and Reynolds number dependent air drag, both improving accuracy of simulations. The computed results demonstrate that the simulation model captures the dynamics of overhead fly casts. As discussed in section 4, results change in agreement with practical experience for changes in how the cast is performed by the caster. Furthermore, it is shown that changes in the tackle e.g., different rods and leaders, produce results in agreement with practical experience. The model is sensitive enough to evaluate even comparatively small tackle changes as demonstrated by small variations in leader configuration. The results also illustrate the energetics of fly casting including kinetic and potential energy components and the work done by external sources (i.e., from the caster's hand and by air drag). These quantities reveal how the energy input from the caster's hand propagates through the fly rod to the fly line, through the loop and ultimately to the leader and fly. Quantifying this transfer of energy increases our ability to understand the dynamics of fly casting.

Declarations

The authors have no competing interests to declare that are relevant to the content of this article.

Ethical statement

The study adhered to the ethical requirements of the host institution.

Contributions

Model development, programming and simulations were performed by Hans Ekander. Theoretical support and experimental data were provided by Noel Perkins. Producing expert cast and assessment of realism in simulated casts were performed by Bruce Richards. The first draft of the manuscript was written by Hans Ekander and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Authors and affiliations

¹ Licentiate of Engineering, Åhus, Sweden

² Professor Emeritus, Mechanical Engineering, University of Michigan

³ Scientific Anglers product developer (retired), FFI Master Casting Instructor

Corresponding author

Correspondence to: Hans Ekander, he@willmanco.se

Supplementary Information

Online resource 1

Online resource 2

Data availability

Data sets generated during the current study are available from the corresponding author on reasonable request.

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